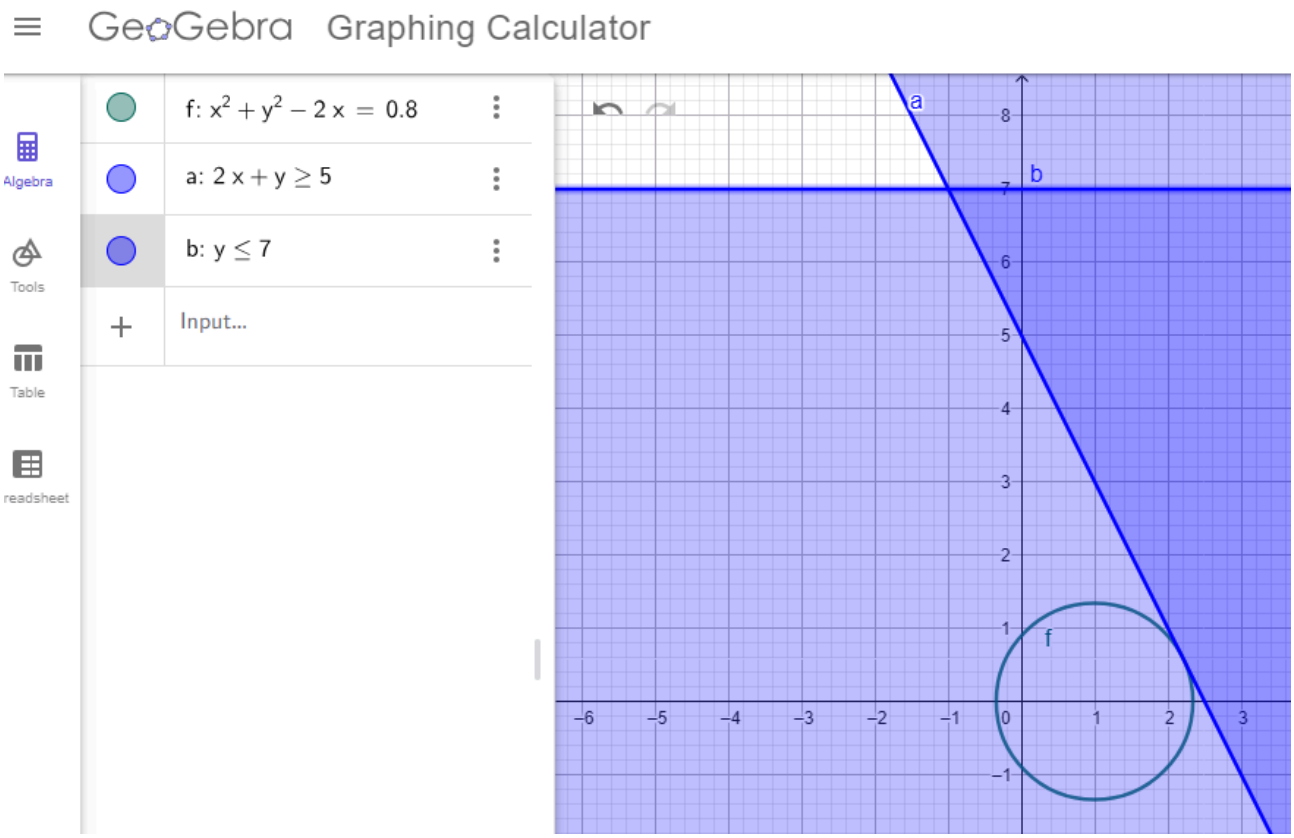
This notebook demonstrate the primal-dual based algorithm.

$$\begin{aligned} \min\ & x_1^2 + x_2^2 - 2x_1 \\ \text{s.t. } & 2x_1 + x_2 \geq 5 \\ & x_2 \leq 7. \end{aligned}$$



Solving the primal problem graphically, we obtain the optimal solution (x^* = 0.8).

Now, let's solve the above problem with primal-dual algorithm.

Lagrangian function: $L(x, \lambda) = x_1^2 + x_2^2 - 2x_1 + \lambda_1(5 - 2x_1 - x_2) + \lambda_2(x_2 - 7)$, $\lambda_i >= 0$

```
import numpy as np
```

```
def f(x):
  return x[0]**2 + x[1]**2 - 2*x[0]
```

```
def L(x,lam):
  return f(x) + lam[0]*(5-2*x[0]-x[1]) + lam[1]*(x[1]-7)
```

```
def gradL_x(x, lam):
  return np.array([2*x[0] - 2 - 2*lam[0], 2*x[1] - lam[0]+lam[1]])
```

```
def gradL_lam(x, lam):
  return np.array([5.0 - 2.*x[0]-x[1], x[1]-7.])
```

```
tp = 0.5
td=0.5
x = np.array([10.,10.])
lam = np.array([0.,0.])
```

```
for iter in range(10):
  # Primal update
  for i in range(50):
    x = x - tp*gradL_x(x,lam)
  # Dual update
  lam = np.maximum(np.array([0.,0.]), lam + td*gradL_lam(x,lam))
  print(f(x))
```

```
0.8000000000000007
0.7999999999999989
0.8000000000000007
0.8000000000000007
0.7999999999999989
0.8000000000000007
0.8000000000000007
0.7999999999999989
0.8000000000000007
0.8000000000000007
```

```
Start coding or generate with AI.
```